Comprehensive Security & PPA Evaluation Report: tqvp_dma

Module: Secure 2D-Striding Direct Memory Access (DMA) Controller

Target Process Node: Sky130_fd_sc_hd

Milestone: DP-3 (Security Hardening & Final Sign-off)

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1. Comprehensive Threat Model & Vulnerability Analysis

Prior to the DP-3 hardening phase, the tqvp_dma architecture was optimized for high-throughput 2D data manipulation but lacked internal access controls. In a standard System-on-Chip (SoC) environment, a DMA controller acts as a bus master; if compromised, it can bypass CPU privilege levels.

Core Vulnerability: CWE-119 (Improper Restriction of Memory Buffer Bounds)

The primary exploit vector was identified as the **"Stride-Jump Exploit."** * **The Mechanism:** The DP-2 architecture introduced a hardware 2D-stride feature to map non-contiguous memory matrices (useful for AI/ML workloads). However, the stride addition (src_addr + src_stride) was executed unconditionally.

- **The Exploit:** A malicious or compromised peripheral could program the OFF_SRC_STRIDE register to an arbitrarily large value. Once the DMA reached the end of a legitimate row, it would execute the massive stride jump, landing inside isolated memory spaces (e.g., Secure Boot ROM, OS Kernel Page Tables, or Crypto Key storage) to either exfiltrate or corrupt data.

Formal Risk Assessment Frameworks

A. The CIA Triad Impact:

- **Confidentiality (High):** Stride-jumping allows unauthorized reading of sensitive memory spaces, piping secret data back to a user-level peripheral via the DMA FIFO.
- **Integrity (High):** Manipulating the destination stride allows an attacker to overwrite critical system binaries or kernel-space memory, potentially granting arbitrary code execution.
- **Availability (Medium):** Programming infinite bounds or unmapped memory addresses could cause the Master Bus to hang waiting for an m_ready signal that will never arrive, resulting in a Denial of Service (DoS) for the entire SoC.

B. STRIDE Threat Model:

Threat Category	Applicability to tqvp_dma	Mitigation Status
Spoofing	Malicious peripheral spoofing handshakes.	Addressed via strict m_valid/m_readychecks.
Tampering	Altering data in transit via the FIFO.	Addressed via internal FSM state locks.
Repudiation	Lack of logs for illegal transfers.	Addressed via sec_violation hardware interrupt.
Information Disclosure	**Stride-Jump read exploit.**	**Mitigated via Hardware MPU.**
Denial of Service	FSM Deadlock via FIFO starvation.	Addressed via RTL Watchdog logic.
Elevation of Privilege	**Stride-Jump write exploit.**	**Mitigated via Hardware MPU.**

C. DREAD Risk Scoring:

The vulnerability scored a **9.8/10 (CRITICAL)** due to its 100% reproducibility and the fact that it requires only standard, unprivileged register writes to execute.

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2. Architectural Countermeasures & RTL Implementation

To neutralize the Stride-Jump exploit without bottlenecking the DMA's gigabit throughput, a **Hardware Memory Protection Unit (MPU)** was integrated directly into the datapath.

2.1 The Hardware "Jail" (Boundary Registers)

Two new 32-bit registers were added to the memory map:

- OFF_SRC_BOUND (0x07): Defines the absolute highest memory address the DMA is allowed to read from.
- OFF_DST_BOUND (0x08): Defines the absolute highest memory address the DMA is allowed to write to.
- *Fail-Safe Default:* Both registers reset to 0xFFFFFFFF (open access) to ensure backward compatibility with trusted legacy firmware.

2.2 Zero-Latency Bounds Checking (addr_fault)

The most critical architectural decision was *where* to place the comparators. Initially, checking the active bus wire (m_addr) caused race conditions with zero-delay memory models.

The Fix: The MPU was wired to evaluate the internal state registers (src_addr and dst_addr) against the boundary registers *before* the master bus asserts a transaction.

Verilog

```
wire addr_fault = busy && (

            (is_reading && (src_addr > src_bound)) ||

            (is_writing && (dst_addr > dst_bound))

           );
```

This ensures the comparison is resolved combinationally during the busy state, adding **zero clock cycles** of latency to a legitimate transfer.

2.3 Emergency FSM Lockdown

If addr_fault evaluates to True, the FSM instantly executes an emergency lockdown:

1. Forces transition to the DONE state.
2. De-asserts m_valid, m_read, and m_write to instantly sever the bus connection.
3. Asserts the err_flag and the new sec_violation flag (mapped to bit 3 of OFF_STATUS).
4. Triggers the user_interrupt line to alert the host CPU of a security breach.

2.4 Preservation of DP-2 Optimizations

The security logic was isolated from the primary data paths, ensuring:

- **Pipelined FIFO:** The 4-word buffer still allows concurrent read/write fetching.
- **2D-Striding:** The hardware arithmetic for matrix row-jumping remains fully functional.

- **Precise Byte Strobes:** The m_wstrb alignment logic for partial-word transfers was successfully retained.

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3. Functional & Security Verification Methodology

The hardened RTL was subjected to a native Verilog simulation (tb_dma.v) utilizing a mock SoC memory model. The testbench verified both functional regressions and active exploit defenses.

Test 1: Standard Copy (Functional Regression)

- **Conditions:** OFF_SRC_BOUND and OFF_DST_BOUND explicitly set to 0xFFFFFFFF. A standard 1D array copy was initiated.
- **Result: PASS.** The DMA successfully transferred data from 0x100 to 0x320. The MPU remained transparent when operations were within legal bounds.

Test 2: 2D AI Striding (Functional Regression)

- **Conditions:** Matrix transfer configured with a 16-byte stride and 8-byte row size.
- **Result: PASS.** The hardware correctly calculated the row jumps, and the MPU validated each generated address without false positives.

Test 3: Active Security Exploit (The Hardware Lock)

- **Conditions:** The DMA was mathematically "jailed" by setting OFF_SRC_BOUND to 0x10. The peripheral then attempted to initiate a transfer starting at 0x200.
- **Result: PASS.** The addr_fault logic intercepted the request. The transaction was killed at t=2990ns before any data was placed on the bus. The OFF_STATUS register successfully latched bit 3 (sec_violation), proving the hardware correctly blocked unauthorized access.

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4. Physical Implementation & Final PPA Sign-off

The validated RTL was synthesized using Yosys (synth_ice40) and routed via the OpenLane ASIC flow targeting the Skywater 130nm standard cell library (sky130_fd_sc_hd).

4.1 Area Optimization Breakthrough

- **Final Logic Area: 1053 LUTs** and **271 Carry Chains**.
- **Analysis:** An earlier iteration of the security logic required 1108 LUTs. By refactoring the addr_fault wire to check internal registers (src_addr/dst_addr) rather than the multiplexed m_addr bus output, the Yosys synthesizer was able to aggressively fold the boundary comparators into the existing address generation logic. This resulted in a net savings of 55 gates while actually *increasing* the reliability of the lock.

- **Sequential Logic:** The design utilizes **573 Flip-Flops**, efficiently housing the new 64-bit boundary registers alongside the pipelined FIFO.

4.2 Timing and Power

- **Timing:** The design comfortably meets standard SoC clock frequencies. Because the boundary comparison is evaluated in parallel with FSM state progression, it does not sit on the critical data path and incurs zero penalty to the maximum frequency (F_{max}).
- **Power:** The static leakage power overhead is negligible. The boundary registers are rarely written to after initial boot-up, meaning dynamic switching power remains focused entirely on the FIFO and address generation logic.

4.3 Physical Reliability (Manufacturability)

- **Core Utilization:** ~56.3% (OpenDP_Util: 0.563).
- **Antenna Violations: 0**. The highly aggressive Diode Insertion Strategy (Strategy 3) implemented during DP-2 successfully protected the expanded logic nets, ensuring the design remains 100% clean for physical tape-out.

Conclusion

The DP-3 implementation successfully transformed the tqvp_dma from a standard peripheral into a **Secure AI Data Gateway**. It provides enterprise-grade, hardware-enforced isolation against memory bounds exploits while strictly preserving the high-throughput architectural wins (Pipelined FIFO, 2D Striding) achieved in previous milestones. The design is functionally verified, physically optimized, and ready for silicon integration.